3 November 2026

To the Editor
Measurement — Elsevier

Dear Editor,

Please consider the enclosed manuscript, “A decision rule without an operating characteristic: the pairwise sum in conformance assessment of multi-channel boresight alignment”, for publication in *Measurement* as a research paper.

The manuscript analyses a decision rule that is used throughout the alignment of multi-channel instruments but has never been examined as a decision rule. Payloads are allocated tolerances one channel at a time and then checked one pair at a time: every pair of as-built deviations must lie within the sum of the two allocations. The paper shows that this check has no operating characteristic in the sense of JCGM 106. Its rejection region is contained in the region the per-channel check has already rejected, so on conforming data its rejection probability is identically zero and no record of passing builds can test it. The same inequality used in the other direction, to set allocations, is optimistic by a sharp constant: allocations must be 15.5 % larger than pairwise sums imply, the worst case being three channels in an equilateral configuration. The paper then maps where the defect can matter and replays a public ten-channel specification (NASA LRO 431-SPEC-000113 Rev A), in which the accounting closes to the arc-second.

I believe the work belongs in *Measurement* rather than in an instrument-design journal because its subject is not a new instrument but the use of measurement data: what a conformity verdict built on measured deviations can and cannot establish, and what evidence would be required to test the rule that produces it. The paper is written in the vocabulary of conformity assessment — acceptance interval, guard band, producer's and consumer's risk, operating characteristic — and it engages the measurement literature on decision rules, including three papers published in this journal and the JCGM 106 framework.

The manuscript is original, has not been published elsewhere and is not under consideration by another journal. It is the work of a single author, who is responsible for all parts of it and declares no competing interests. The work was supported by the National Natural Science Foundation of China (grant no. T2600787) and the Natural Science Foundation of Shanxi Province (grant no. 202403021212106). Code, transcribed data and derived outputs accompany the submission as supplementary material.

Thank you for your consideration.

Yours sincerely,

Zhengkang Zuo
Polar-vision Laboratory, College of Geoscience and Surveying Engineering, Taiyuan University of Technology
Taiyuan 030024, Shanxi, China
zuozhengkang@tyut.edu.cn — ORCID: 0000-0003-1255-6332